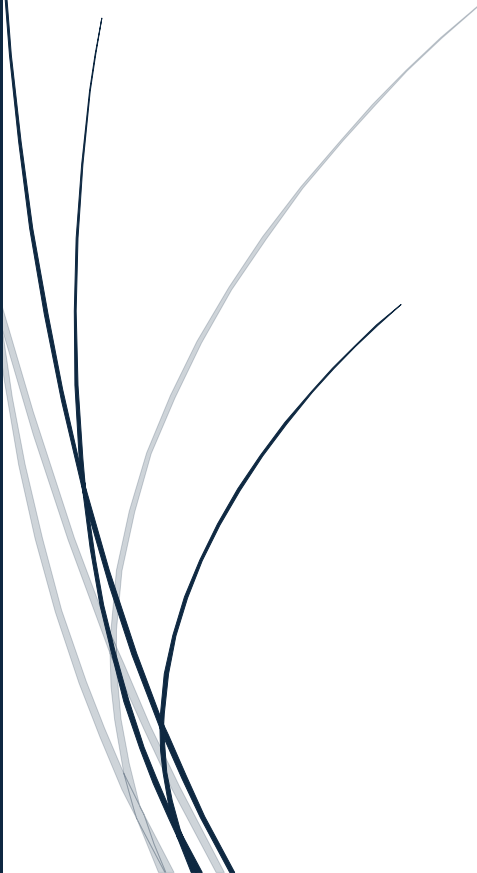


Semester 1, 2025

EGB241 – DC Motor Report

Group 34



David Bui, Josh Cook, Thomas Cox
QUT

Motor Design Proposal

The objective of this design is to construct a reliable and efficient DC motor suitable for home appliances which operate below 30V and 5 amps to ensure low power consumption seen in typical electrical appliances (<150 Watts). The rotor features an 8-slot configuration composed of 16 ferromagnetic lamination plates each 1.2mm thick, equating to a total length of 19.2mm. It operates using a brushed commutated system with electrical connections made through copper segments. The coils were wound from enamelled copper wires configured in a fully pitched lap winding arrangement. Additionally, the rotor is enclosed in a stator assembly consisting of two permanent magnets arranged with a north and south (N-S) polarity. The motor was designed to support bi-directional rotation, to ensure robust performance when used in home applications.

The coil windings were done manually due to the simplicity of the design and its low power applications. A total of eight coils were formed using the 8-slot configuration arranged in two layers of four coils each and were restricted to 45m total length due to resource limitations. However, the number of turns calculated was slightly above the achieved number in practice as it did not account for the overlap between each successive coil formed as well as the low packing factor seen in the inaccuracies from manual windings. Overall, the final designed had utilized 35 turns in total.

Operating Torque	0.0409 Nm
Speed	1695 RPM
Mechanical Output Power	7.25 Watt
Theoretical Efficiency	2.41%

Several sources of loss were not accounted for during theoretical calculation of DC motor parameters, and include:

- ❖ Mediocre packing factor from manual coil winding, causing efficiency losses
- ❖ Airgap between rotor and poles weakens magnetic field induced on coils
- ❖ Winding overhang extending past active magnetic region, not contributing to production of torque
- ❖ Potentially > 50% flux leakage from pole length differing from rotor thickness
- ❖ Brush-to-commutator electrical contact loss
- ❖ Additional frictional loss from brush-to-commutator contact
- ❖ Mechanical loss because of air resistance

It is also important to consider that the motor will operate connected to a loaded dynamometer during the practical demonstration. This will produce different results compared to theoretical calculations due to varied load conditions. Because of this and the previously listed losses, it is difficult to accurately approximate the real behaviour of the system.

Design Restrictions

The design and construction of the motor was strictly limited to these constraints following the client's (QUT) expectations:

1. Motor must be **brushed and commutated**
2. Motor may **not draw more than 5A** and **operate less than 30V**
3. No load speed limited to **2000 RPM**
4. Motor must overcome internal resistance to rotation of the Dynamometer (**0.000845Nm**)
5. Ideally **start without external assistance**
6. Field must be supplied by **TWO permanent magnets** (with or without pole shaping)

Part A – Design Phase

Material Selection:

Before the construction phase, three key inputs were considered to determine electromagnetic and mechanical parameters of the motor. These inputs were:

1. Wire gauge
2. No. of turns per coil
3. Magnet Selection

Table 1 outlines the range of wire gauges available to be used in the copper windings in each coil. The wire thickness was chosen based on a selection criterion that best suited the design intentions as outlined in the motor proposal. As the motor is intended for home appliances it should be durable, long-lasting and made to withstand prolonged uses induced by heating. Therefore, a thicker wire was chosen due to its lower resistance per meter compared to thinner wires. A lower resistance means the wire is less susceptible to heating during prolonged operations which improves the motor's power rating. In contrast, thinner wires were found to have higher resistance, making them less suitable for sustained uses due to increased heating and reduced current carrying capacity.

Further considerations were made to ensure that the wire was not too thick, as this would reduce the number of turns that could fit in the rotor slots. Since the coil windings are done manually, the resulting packing factor is expected to be lower than that of industrially wound coils, which typically achieves a much tighter and more compact winding arrangement. As a result, a 21 gauge was selected for the coil windings with a 0.723 mm thickness in diameter and 12.8mΩ of resistance per meter.

Table 1: Selection of Available Wire Gauges

AWG	Diameter	Turns of wire, without insulation	Area	Resistance/Length
	(mm)	(per cm)	(mm ²)	mΩ/m
18	1.024	9.77	0.823	6.385
19	0.912	11	0.653	8.051
20	0.812	12.3	0.518	10.15
21	0.723	13.8	0.41	12.8
22	0.644	15.5	0.326	16.14
23	0.573	17.4	0.258	20.36
24	0.511	19.6	0.205	25.67
25	0.455	22	0.162	32.37
26	0.405	24.7	0.129	40.81

27	0.361	27.7	0.102	51.47
28	0.321	31.1	0.081	64.9
29	0.286	35	0.0642	81.84

After selecting the desired wire gauge, the number of turns per coil was calculated based on the geometric parameters outlined in Table 2. The total length of wire used for a single turn was determined by summing the lengths of wire inside the rotor slots and the wire outside the slots:

$$\begin{aligned}
 \text{Wire inside slot} &= 2 * L_R \\
 \text{Wire outside slot (chord length, } L_c) &= 2 * r_{ave} \\
 \text{Length of One turn (} L_{W1}) &= \text{Wire inside slot} + \text{Wire outside slot}
 \end{aligned}$$

The wire sitting inside the rotor slot corresponds to twice the rotor lamination length as each coil spans across two slots. Likewise, the chord length (L_c) represents the portion of the coil that lies across the surface of the laminations, that is outside the slots.

The total length of one coil is calculated as:

$$\text{Length of One coil} = L_{W1} * N_c$$

Where N_c is the number of turns.

The total length of copper wired used for the entire winding is then:

$$\text{Total length (} L_W) = \text{Length of One Coil} * \text{No. of Coils}$$

These geometric properties were adjusted by changing the no. of turns such that the total copper wire used did not exceed 45 m according to the design constraints. The final values consisted of a total of 43 turns per copper wire with approximately 44m used. The remaining copper length provides a safety margin to account for uncertainties caused by manual winding, low packing factors and human error.

Table 2: Theoretical Coil Measurements Based on Selected Number of Turns

Length of rotor (L_R):	19.2	mm
Radius of Windings (r_{ave}):	22.5	mm
No. of turns of wire per coil (N_c):	43	
Chord Length (L_c):	45	mm
Length of 1 turn (L_{W1}):	128.4	mm
Length of 1 coil:	5521.2	mm
Length of all coils (L_W):	44.1696	m (< 45 m)

The magnets were chosen out of a selection of four rare-earth metals, each with varying magnetic flux densities as outlined in Table 3. The magnet #3495 was ultimately selected as it provided the highest magnetic flux possible and therefore was the strongest magnet available. This decision was guided by these key mechanical formulas:

$$Torque_{induced} = K_2 * \phi * I, \text{ where } K_2 \text{ is a constant}$$

$$RPM_{max} = V_T * \frac{60a}{ZP\phi}$$

As shown, the motor's induced torque rises linearly with the magnetic flux (ϕ) whilst the maximum rotational speed decreases inversely with flux. Since the motor was designed to operate under loading conditions, maximum torque generation was prioritized over speed instead due to the design constraints specifying that at no-load the rotational speed should not exceed 2000 RPM, making high rotational speed a secondary priority. A higher torque ensures the motor can overcome any external and mechanical loads without stalling such that it provides a higher standard of efficiency.

Table 3: Selection of Available Magnets

AVAILABLE MAGNETS						
CODE	B (G)	LENGTH (L) (mm)	WIDTH (W) (mm)	HEIGHT (mm)	Surface Area (cm ²)	ϕ (mWb)
#2522	1400	50	20	12.7	10	0.14
#3460	2408	25	12.5	3.5	3.125	0.0753
#3470	3625	25	12.5	6	3.125	0.1133
#3495	3709	40	15	8	6	0.225
LM1614	580	20	15	5	3	0.0174

Excel Analysis of Field and Armature Windings:

The properties include the motor constants K1 and K2, armature resistance (Ra), terminal voltage, maximum speed and torque, as well as the electrical and mechanical power to determine the overall efficiency, were all calculated within excel to then be used in MATLAB to simulate the motor behaviour. However, the scope of this spreadsheet is limited to evaluating the motor's performance metrics under **no-load conditions and without the dynamometer coupling**. As such, the performance metrics reflect **ideal operations** and do not account for external losses induced by loads. In contrast, the MATLAB simulations will replicate a more practical scenario that introduces both internal friction of the motor and the resistive torque applied by the dynamometer.

Magnetic Properties:

The #3495 magnet that was selected was evaluated under the assumption that there will be a 50% flux leakage meaning only half of the magnetic flux will effectively interact with the armature windings. By using a conservative baseline, it accounts for the low packing factor induced from manual windings and the unavoidable air gaps that will be present. This assumption helps to validate the motor's performance under realistic and potentially suboptimal conditions which is typically seen in real-world scenarios. As a result, the total flux that passes through the rotor is dependent on the following formulas:

$$\phi_{total} = B * A$$

$$\phi_{linkage} = \phi_{total} * 0.50$$

Where:

- "B" is the magnetic flux density in Tesla (converted from Gauss as shown in Table 3)
- "A" represents the surface area of the magnet facing the armature windings.

As shown in Table 4, the total flux interacting with the armature winding is half of the total flux produced by the magnet equating to an effective flux of 0.106 mWb. This value is used in the MATLAB simulation for the final motor.

Table 4: Magnetic Properties of Magnet #3495

Parameter	Value	Unit
Flux density at surface (B):	3709	G
Area of magnet surface (A):	6	cm ²
Total flux of magnet (ϕ_{Total}):	0.212625806	mWb
Flux passing through rotor ($\phi_{linkage}$):	0.106312903	mWb

Electromagnetic Constants / MATLAB Inputs:

Key parameters were computed that govern the electrical performance of the DC motor that will be used in MATLAB simulations. These were the motor constants K1 and K2 and the armature resistance of the copper windings.

Both the constants K1 and K2 are fixed values that can be found by:

$$\text{Voltage Constant}(K_1) = \frac{ZP}{60a}$$

$$\text{Torque Constant}(K_2) = \frac{ZP}{2\pi a}$$

$$\text{where } Z = 2 * C * N_c$$

The value Z represents the total number of conductors in the armature windings conductors in the armature windings – this depends on the total number of coils and turns used. The number of poles, “P”, is fixed to 2 since the motor uses two permanent magnets mounted in the stator. Given the motor’s lap winding configuration and a 2-pole design, the number of parallel paths, denoted as “a” is also fixed at 2.

In this configuration, the winding structure implies that 4 coils are wound in series to form one parallel path (1st layer), while another 4 coils will be wound in series to form the second parallel path (2nd layer). Both layers are connected in parallel forming the complete lap winding arrangement.

A 1Ω resistance is assumed between the brush and commutator to account for contact resistance and other interface losses.

The total armature resistance is then calculated as follows:

$$\text{Armature resistance } (\Omega) = 1 + \frac{\text{total wire length} * \text{resistivity of selected gauge}}{4}$$

The division of 4 accounts for the process of isolating one series path – which is half the total wire length – and then calculating the equivalent resistance in parallel. Table 5 outlines the final motor constants used in the MATLAB simulations.

Table 5: Motor Constants Used in MATLAB Simulations

Motor Constant (K₁):	11.46666667
Motor Constant (K₂):	109.4986008

Electrical and Mechanical Properties:

Table 6 outlines the minimum terminal voltage and maximum armature current under no-loading conditions. A terminal voltage of 2.4 allows for a maximum rotational speed of approximately 1969 RPM which satisfies the design constraint of remaining under 2000 RPM.

Table 6: Electrical and Mechanical Characteristics of the Motor When Rotating

Terminal Voltage (V_T):	2.4	V
Maximum Armature Current ($I_{a,max}$):	1.639589753	A
Maximum Rotational Speed ($n_{r,max}$):	1968.73869	rpm
Maximum Generated Torque (T_{max}):	0.019086651	Nm

Furthermore, the motor must be able to generate a maximum torque greater than the dynamometer's stalling torque to ensure it can start rotating. The dynamometer torque was determined to be 0.000845Nm, while the calculated maximum torque of the motor is approximately 0.019Nm, which comfortably exceeds this requirement as does the current draw measuring 1.64 A which is well below the 5A limit indicating a promising electrical performance under no load. However, it is important to acknowledge that these parameters will change when the motor is subjected to external loads and torque.

Expected Speed and Torque graphs

An evaluation of all available magnets identified #3495 to be most optimal, based on superior torque performance in DC motor application. The highest torque reading ensures the best motor performance is achieved for the system. It is evident that with the limitation of a 5-amp current draw (operating at 7.2 volts, 4.9 Amps), the DC motor will not apply enough torque to even spin the rotor under no-load conditions with #2522, #3460 and #3470 magnets (Figure 1). This is a result of the magnets not being able to induce the no-load torque requirement of 0.040845 Nm. Therefore, the only viable magnet was #3495.

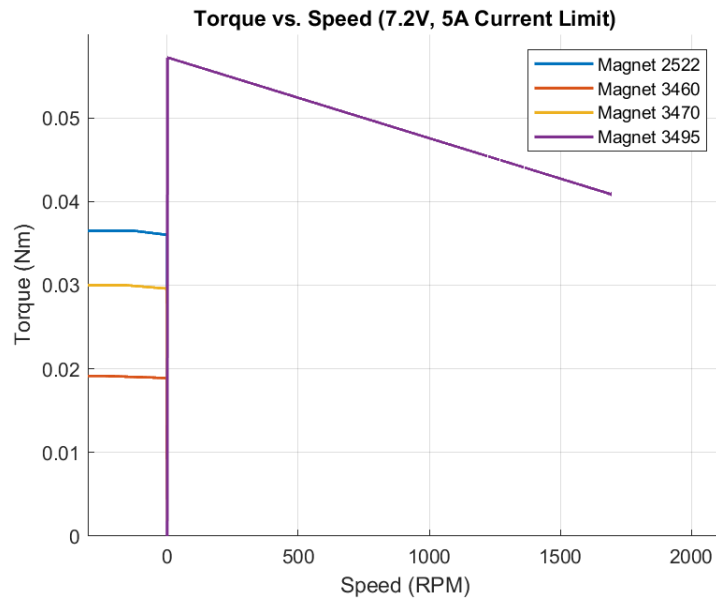


Figure 1: Torque vs Speed curve for each magnet

A secondary graph was created without the limitation of current draw (increased voltage to 20V) to determine the behaviour of each magnet during motor operation resulting in #3495 producing the highest torque at lower RPM due to its greater magnetic induction (3709 Gauss). This magnet also has a steeper torque drop-off, eventually producing less torque than other magnets at extremely high speeds. Motors with weaker magnets that produce less back EMF maintain more torque at higher speeds. Such a consideration is negligible for the speed limitation of 2000 RPM. Even if it were an outstanding issue, this magnet would have still been chosen as it is the only magnet that will work under the criteria limitations. Subsequently, the viability of #3495 still holds.

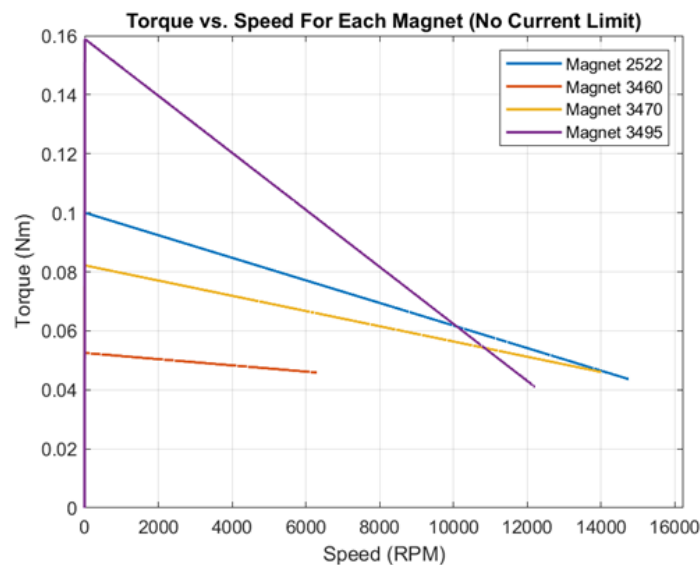


Figure 2: Labelled coils to be placed on commutator

One other factor to consider is the dimensions of the magnet. #3495 may have the highest magnetic flux, but its dimensions may be suboptimal relative to the rotor's thickness. This suboptimality comes in the form of potential flux leakage because of a 40:19.2 mm magnet length to rotor thickness ratio. Theoretical calculations already accounted for a 50% flux leakage within the system, but considering this ratio difference, #3495 may have more adverse effects. Nevertheless, this dimensional mismatch must be tolerated, as the plot clearly indicates that other magnets are not viable.

The exponential form of speed vs time (Figure 3a) is expected and comes from torque balance and torque EMF feedback acting as a damping effect. Essentially, the motor takes time to overcome frictional load before stabilizing. A 1695 RPM maximum speed ensures the motor is well within the 2000 RPM limit at no-load. Although stabilization occurs after approximately 18 seconds, it is not a problem for this application.

The function depicted within figure 3c displays its maximum efficiency occurring at half of the maximum RPM. 2.41% efficiency could be regarded as very low but is a result of armature resistance and small physical size creating high relative losses. The negative exponential displayed in figure 3b is expected, as the building speed increases back EMF which reduces system net current. A subsequent decrease in current reduces torque.

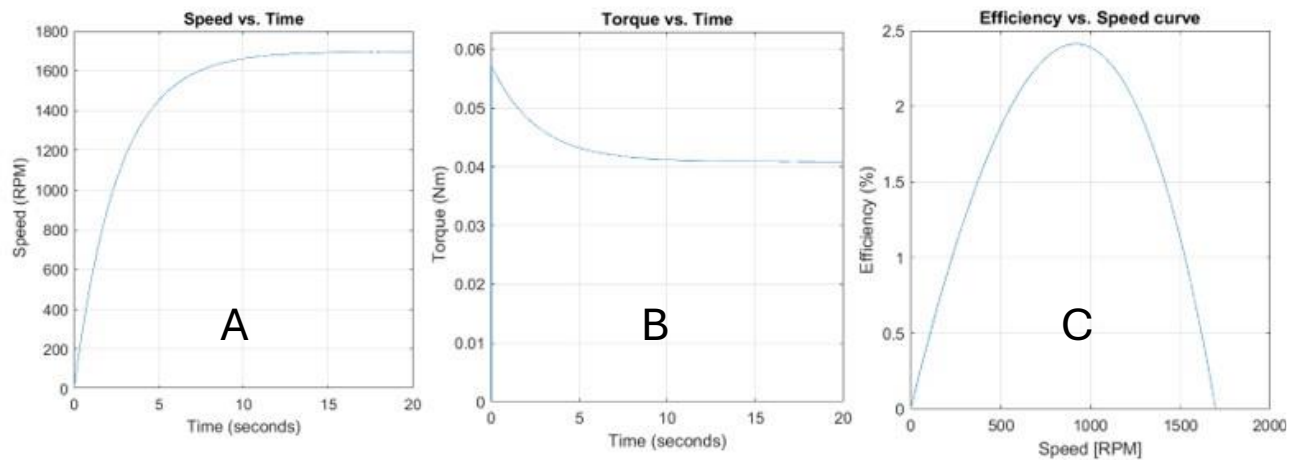


Figure 3: Mechanical characteristics over time

Acceleration Graphs

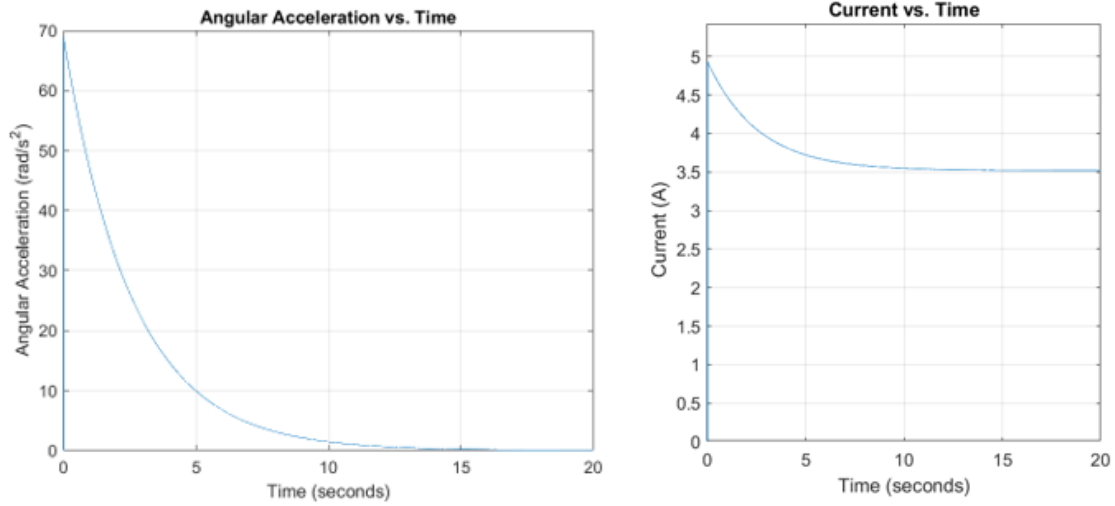


Figure 4: Acceleration and Current with NO LOAD

Table 7: Key Parameters Obtained Under No Load

Supply Voltage	7.2 V
Minimum current drawn (no-load)	3.5196 A
Maximum current draw (no-load)	4.9304 A
Time to reach maximum speed	17.7598 seconds

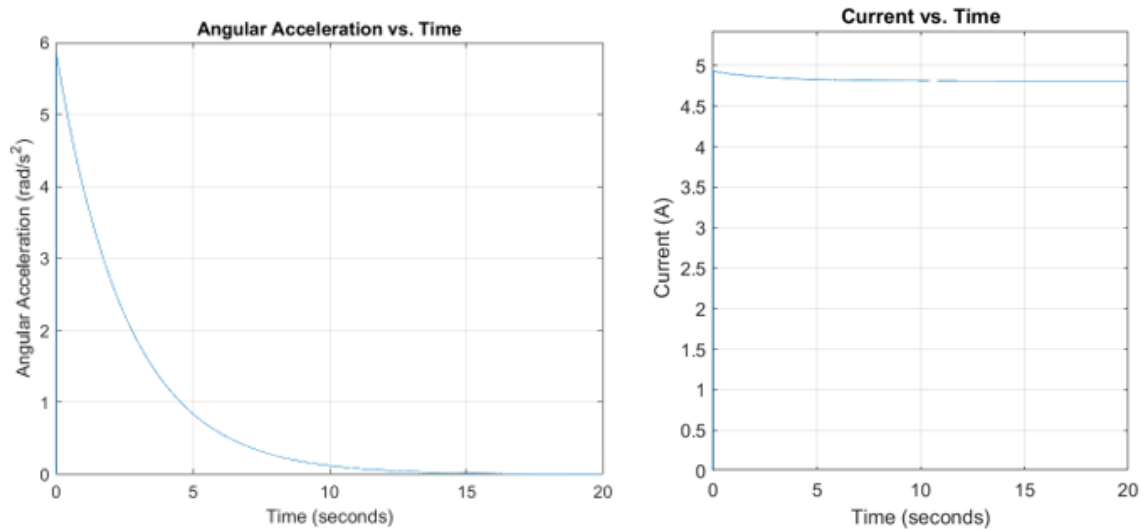


Figure 5: Acceleration and Current graph WITH LOAD

Table 8: Key Parameters Obtained Under a Load

Supply Voltage	7.2 V
Minimum current drawn (0.015 Nm load)	4.8114 A
Maximum current draw (0.015 Nm load)	4.93114 A
Time to reach maximum speed	15.768 seconds

Undergoing comparative analysis for no-load and load conditions, it can be determined that key performance differences occur due to the influence of induced load on the system. Under no-load (Table 7) there is a larger difference between minimum and maximum currents as opposed to the system under 0.015 Nm load (Table 8) (1.41 Amps compared to 0.11974 Amps respectively). This occurs because, under no-load conditions, the motor accelerates to a higher speed, generating greater back EMF which significantly reduces steady state current. In contrast, the applied load limits acceleration and motor speed, resulting in lower back EMF and a smaller drop in current from startup to steady state.

Part B – Construction and Testing

Iterative Testing and Processes

To gain a clear understanding of each component's function and optimize the performance, an iterative process of construction and testing was implemented. Before constructing the first iteration, the rotor was prepared by shrink-wrapping sharp edges to prevent damage to the windings, and a lap-winding technique was used to layer the copper wire as much as possible. This resulted in a total winding count of 35 per coil as opposed to the calculated 43. To avoid confusion between the coil windings, labels were taped to the end of each one which specified which section of the commutator they would be attached to in future (Figure 6).



Figure 6: Labelled coils to be placed on commutator



Figure 7: Commutator segments spaced 2mm apart

The commutator consisted of eight copper strips evenly spaced at 2mm and attached to a plastic cylinder via an adhesive (Figure 7). These were labelled from 1-8 to track the placements of the corresponding coil onto the correct commutator segment. Given the configuration of the rotor, each slot was equally separated by 45 degrees. To achieve maximal torque generated by the interaction between the coil windings and magnet field, the coils should be positioned 90 degrees from their corresponding commutator segments as described by Lorentz's law. Therefore, the alignment of the coil and commutator was easily achieved by rotating the commutator by two slots to ensure the correct angular offset for optimal force production. The ends of the copper wiring were then soldered directly to these strips.



Figure 8: Brush contact with commutator during assembly

Once the assembly of the windings and commutator segments were complete, the rotor was placed within the acrylic chassis in which, it was placed directly between the two magnets that had opposing poles facing it. These magnets were attached to the chassis via screws and nuts and were pushed closer to the rotor via the use of wooden spacers. To conduct the flow of electricity into the commutator, brass strips were hung from above to run tangentially to the commutator and were slightly bent to minimize the contact area such that the commutator could rotate through the coils without overlapping two coils at once and creating a short (Figure 8).

With these components in place, an initial test was conducted using **only a power source** under **no loading conditions**. The purpose of this is to assess whether it would function at a basic level in an ideal scenario as demonstrated in the excel. After some adjustments such as rotating the commutator to the appropriate 90° in relation to the rotor, as well as ensuring the brass strips did not contact multiple coils at once, the machine shown in Figure 9 successfully spun at approximately 2.1V. This value closely reflects the 2.4V measured in the excel spreadsheet (Table 6) which indicated a promising iteration.

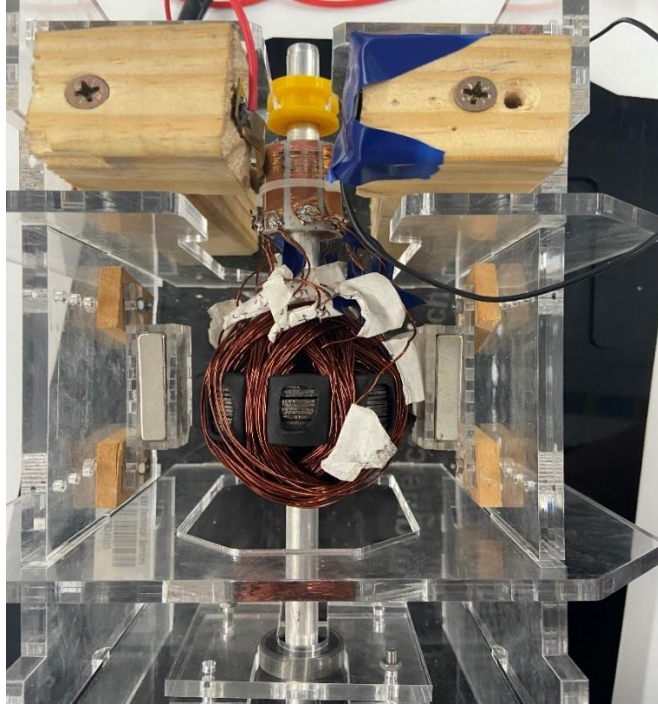


Figure 9: Assembled Initial iteration

With the initial iteration complete and functional, it was then attached to a dynamometer to measure the rotational speed (RPM) as well as the output voltage and current. The load dial was gradually increased to test for efficiency; however, the motor would stop spinning once this load was applied regardless of how much voltage was inputted. To fix this, the machine was disassembled to assess individual components for troubleshooting. What was found was that the lack of torque for the motor to start was a combination of four primary issues:

- Friction from loose tape and shrink wrap
- Loss of electrical continuity
- Spacing magnets resulting in large air gaps
- Unstable magnet casing

To reduce friction, the shrink wrap was filed down such that there were no parts poking out, the taped labels were removed to ensure it did brush against the magnet, and the loose wires were carefully compressed into the rotor spacings by clamping with a vice.

To further investigate the loss of electrical continuity a multimeter was used to measure the resistance across each copper commutator segments. A few segments had displayed very high resistance values in the megaohm range which indicated minimal to no electrical conductivity in those areas. In contrast, other segments showed lower resistance readings around 1 ohm suggesting the presence of electrical flow in those areas. It was discovered that the commutator segments with high resistance values corresponded to areas where the enamel coating on the copper coil had not been adequately stripped resulting in excessive insulation and therefore poor electrical conductivity.

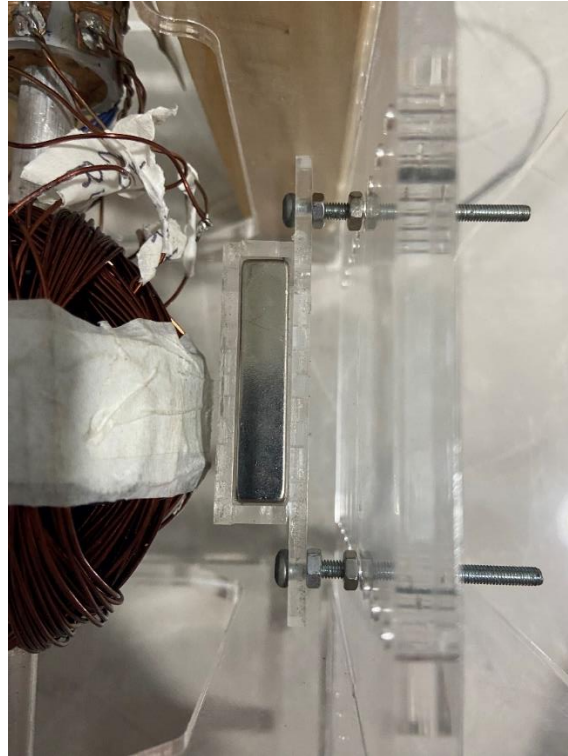


Figure 10: Magnet spacing casing using nuts and bolts to rectify stability issues

Finally, the issue with the magnet spacing and stability issues were addressed by inserting a 40mm screw fixated by three nuts. As shown in Figure 10, three nuts were strategically placed along the screw shaft to provide consistent axial alignment. Two nuts are positioned inside of the chassis which set the distance of the magnet casing from the motor. These were spaced apart until a reasonable amount of pulling torque from the magnet was felt when spinning the rotor itself. The final nut lays outside of the chassis which tightens and locks the screw in place ensuring a stable arrangement. This was more stable than the previous design using the wooden supports because the nut and screw assembly offers greater mechanical rigidity when fixating to the side of the chassis. In contrast, the wooden supports were prone to flexing and loosened gradually due to vibration from the rotor spinning.

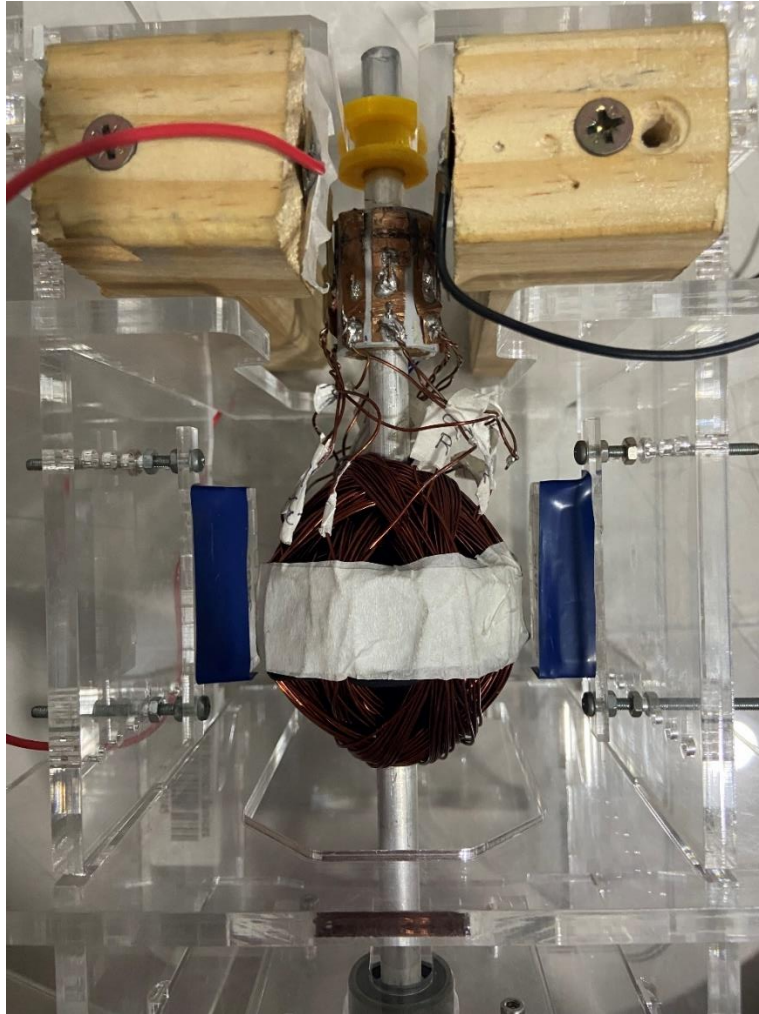


Figure 11: Final assembly of motor

The final iteration of the motor (Figure 11) consisted of coil windings that were tapped down and appear much tighter compared to the loose arrangement seen in Figure 9. The coils were carefully arranged in a 90-degree configuration as evidenced by the twisting of the coils and each are soldered onto the commutator strip. The magnets were also stabilized with three screws and taped down to minimize collision risk. During testing the design was able to rotate under the dynamometer load. However, the limited durability of the copper strip as seen in Figure 11 resulted in the breakdown of the commutator at high-speed testing.

Output Results for DC Motor

Readings were not obtained during practical demonstration due to damaged / burnt commutator strips creating an open circuit in the coils. This issue occurred just before activating the load after it reached nearly 2000 RPM, consequently ceasing operation. Fortunately, previous output readings were obtained when the commutator was in better condition:

Speed	923 RPM
Power Input	43.002 Watts
Power output	0.0390 Watts
Efficiency	0.0906 %
Output Torque	0.403 mNm

Output values are evidently poor because of additional load and previously illustrated factors contributing to loss within the real system. This DC motor is not ideal for practical applications as evident from its poor efficiency and output torque.

Review Phase

Teamwork Breakdown:

Our team had divided responsibilities based on individual strengths as well as each member's personal commitments. This approach helped us distribute the work fairly while ensuring that each task was handled by the most appropriate team member given their circumstances. For example, **Thomas** was more hands-on and had access to a personal workshop and the tools required to build the motor. As a result, he was primarily assigned to handle the physical assembly of the DC motor, such as cutting the wooden supports and fabricating other components required to stabilize the components in the chassis.

In contrast, **Josh** and **David** live far from the university, which makes frequent in-person testing difficult. Therefore, they were responsible for majority of the theoretical aspects of the design such as developing the Excel Spreadsheet and running MATLAB simulations.

Despite the division of responsibilities, each team member contributed to the motor's construction and helped with the other person's section when necessary. For instance, Josh demonstrated exceptional skill in coil winding. His ability to maintain consistent tension during manual winding allowed us to achieve the maximum number of turns without coils becoming loose or overhanging from the rotor.

Analysis of how concurrent engineering modified thought process and outcome:

Concurrent engineering is a methodology where the design and development stages of a project occur simultaneously rather than sequentially. Traditionally, a single person may follow a linear workflow – they design first, build and then test. However, we found this approach to be inefficient use of our time. By adopting the principles of concurrent engineering, it encourages us to work more collaboratively and have a parallel workflow.

For example, at the beginning of the project, our focus was primarily on the theoretical aspects of the design – fine tuning the electromagnetic parameters to achieve an idealized performance. However, concurrent engineering prompted us to consider practical constraints alongside the theory. One such consideration was whether the number of windings calculated in our Excel spreadsheet would physically fit within the rotor slots. Instead of waiting on one person to finalize the winding calculations, we worked in parallel: while one member refined the MATLAB simulations and the spreadsheet, another experimented with different numbers of turns to assess its practical feasibility. This simultaneous approach improved our work efficiency by aligning and adjusting our theoretical parameters with the physical constraints of the system early in the process.

Through this parallel workflow we were able to efficiently form multiple design iterations which allowed us to fine tune our physical parameters in a timely and coordinated manner.

Recommendations and Future Improvements:

While our DC motor was successful at demonstrating its ability to rotate under loading conditions, there were several areas where improvements could be made. One notable issue that we encountered was the low durability of the copper commutator strips. During successive testing the strips would frequently burn off due to heat and friction from high rotating speeds which required us to repeatedly replace them. This was highly inefficient in regards to both time and money. For our future designs, we would use more durable material, such as solid copper bars to improve the commutators' longevity.

Additionally, we noticed that during testing there was substantial friction between the brushes and commutator. This high friction was caused by poor contact placement between these surfaces which resulted in significant arcing and sparks. Consequently, we observed this had caused a higher voltage and current draw compared to what was calculated, as well as causing the brass material to have burn marks. In future iterations of the motor, we would substitute the brass material for carbon instead, as this material has a smoother contact surface which would reduce material wear. Furthermore, we would improve the contact design by using rounded brush tips to minimize the contact area and by incorporating springs to adjust the brush displacement during testing. These changes could help improve the motor's longevity, reduce friction and power consumption.